67978: A Needle in a Data Haystack

Introduction to Data Science

Homework #1 — Practical Part: Consolidated Report

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1. Crawl Summary

Source site	https://www.bookdelivery.com/
Top-level categories crawled	20 (target: all top-level categories)
Pages per category (max)	5
Total books collected	3,542
Median Price in NIS (used for IsExpensive split)	74.35
NIS → USD exchange rate	3.01 (per spec)
Polite delay between requests	3 seconds
Retry policy	3 attempts with exponential backoff
Tools	Python, requests, BeautifulSoup, undetected-chromedriver (for Cloudflare bootstrap), pandas

Pipeline (Steps 1–5)

Step 1 (Crawl). Discover all category links from the homepage programmatically by parsing the *inyectaSubCategorias* JSON in the homepage script. For each category, follow up to 5 pagination pages, extract individual book URLs, and visit each one. Per book extract Title, Categories, Authors, Price (NIS & USD with math.ceil rounding to 2 decimals), Year, Synopsis, NumberOfReviews, StarRating ("None" for zero reviews), Language, Format, Dimensions, Weight and ISBN/ISBN13.
Step 2 (DataFrame). Load every record into *df_books*, cast numeric fields, write *books_raw.csv*, *books_raw.json*, plus a single-record *books_example.json* alongside *books_example.jpg*.
Step 3 (Preview / Sort). Save the first 10 rows before sorting, then again after sorting by Title ascending.
Step 4 (Derived features). Add *IsExpensive* (price > median NIS) and *NumberOfAuthors* (comma split). Save the processed DataFrame and an updated JSON.
Step 5 (Summary). Compute mean / std / min / max / median for Price in USD, Year, StarRating, NumberOfReviews and NumberOfAuthors over non-missing values, plus total row count.

2. First 10 rows — before sorting

Showing the most informative columns. The full 20-column schema (including Synopsis text, Language, Format, Dimensions, Weight, ISBN/ISBN13 and url) is preserved in the CSV/JSON files in output/.

id	Title	Category	Authors	Price in NIS	Price in USD	Year	StarRating	NumberOfReviews	Synopsis length
1	Consecration to St. Joseph: The Wo…	Biography,Literature …	MIC Donald Calloway	74.41	24.73	2020	4.5	2	384.0
2	Everybody Is a Poem: Midlife in Rh…	Biography,Literature …	Jan Brierton	41.75	13.88			0	619.0
3	The Power Broker: Robert Moses and…	Biography,Literature …	Robert A. Caro	74.62	24.80	1975	5.0	3	3478.0
4	what remains,a memoir of fate, fri…	Biography,Literature …	Radziwill, Carole	70.11	23.30			0	1264.0
5	Moonwalk - Michael Jackson	Biography,Literature …	Michael Jackson	81.13	26.96	2009	4.91	11	1483.0
6	Dancing at the Edge of the World: …	Biography,Literature …	Ursula K. Le Guin	56.76	18.86	1997		0	550.0
7	The Year of Magical Thinking - Jo…	Biography,Literature …	Joan Didion	48.18	16.01	2007		0	1290.0
8	"Surely You're Joking, mr. Feynman…	Biography,Literature …	Richard P. Feynman	55.27	18.37	2018		0	618.0
9	The Billionaire who Wasn't: How Ch…	Biography,Literature …	Conor O'Clery	96.75	32.15	2013		0	1093.0
10	Napoleon: A Life - Roberts, Andrew	Biography,Literature …	Andrew Roberts	78.96	26.24	2015	5.0	3	1616.0

Source: *output/books_before_sort.csv*

3. First 10 rows — after sorting by Title

Showing the most informative columns. The full 20-column schema (including Synopsis text, Language, Format, Dimensions, Weight, ISBN/ISBN13 and url) is preserved in the CSV/JSON files in output/.

id	Title	Category	Authors	Price in NIS	Price in USD	Year	StarRating	NumberOfReviews	Synopsis length
8	"Surely You're Joking, mr. Feynman…	Biography,Literature …	Richard P. Feynman	55.27	18.37	2018		0	618.0
760	$100M Offers: How to Make Offers s…	Economics,Finance,Bus…	Alex Hormozi	104.69	34.79		5.0	2	3310.0
1858	(n).concise oxford thesaurus.(3a.e…	Language And Linguist…		76.42	25.39			0	
1742	(n).shorter oxford english dict.(+…	Language And Linguist…	Oxford Languages	354.75	117.86	2008		0	
3463	1,500 Color Mixing Recipes for Oil…	The Arts	Powell, William F.	69.92	23.23	2012	5.0	1	1006.0
939	100 Cars That Changed the World: T…	Economics,Finance,Bus…	Publications Internat…	75.28	25.01	2020		0	633.0
3123	100 Deadly Skills: Combat Edition:…	Society And Social Sc…	Emerson, Clint	75.53	25.10	2021		0	1956.0
3135	100 Deadly Skills: Survival Editio…	Sports And Active Out…	Emerson, Clint	68.28	22.69	2016	5.0	1	1537.0
3049	100 Deadly Skills: The Seal Operat…	Society And Social Sc…	Emerson, Clint	55.99	18.61	2015		0	251.0
2284	100 Dot-to-Dot Puzzles Book: Jumbo…	Lifestyle,Hobbies And…	Happiness Dots	65.73	21.84			0	

Source: *output/books_after_sort.csv*

4. First 10 rows — after processing

Now includes the derived **IsExpensive** and **NumberOfAuthors** columns.

id	Title	Category	Authors	Price in NIS	Price in USD	Year	StarRating	NumberOfReviews	Synopsis length	IsExpensive	NumberOfAuthors
1	Consecration to St. Joseph: The Wo…	Biography,Literature …	MIC Donald Calloway	74.41	24.73	2020	4.5	2	384.0	1	1
2	Everybody Is a Poem: Midlife in Rh…	Biography,Literature …	Jan Brierton	41.75	13.88			0	619.0	0	1
3	The Power Broker: Robert Moses and…	Biography,Literature …	Robert A. Caro	74.62	24.80	1975	5.0	3	3478.0	1	1
4	what remains,a memoir of fate, fri…	Biography,Literature …	Radziwill, Carole	70.11	23.30			0	1264.0	0	2
5	Moonwalk - Michael Jackson	Biography,Literature …	Michael Jackson	81.13	26.96	2009	4.91	11	1483.0	1	1
6	Dancing at the Edge of the World: …	Biography,Literature …	Ursula K. Le Guin	56.76	18.86	1997		0	550.0	0	1
7	The Year of Magical Thinking - Jo…	Biography,Literature …	Joan Didion	48.18	16.01	2007		0	1290.0	0	1
8	"Surely You're Joking, mr. Feynman…	Biography,Literature …	Richard P. Feynman	55.27	18.37	2018		0	618.0	0	1
9	The Billionaire who Wasn't: How Ch…	Biography,Literature …	Conor O'Clery	96.75	32.15	2013		0	1093.0	1	1
10	Napoleon: A Life - Roberts, Andrew	Biography,Literature …	Andrew Roberts	78.96	26.24	2015	5.0	3	1616.0	1	1

Source: *output/books_processed_preview.csv*

5. Summary statistics

Computed only over books where the column has a value. *total_rows* is the row count of *books_processed.csv*.

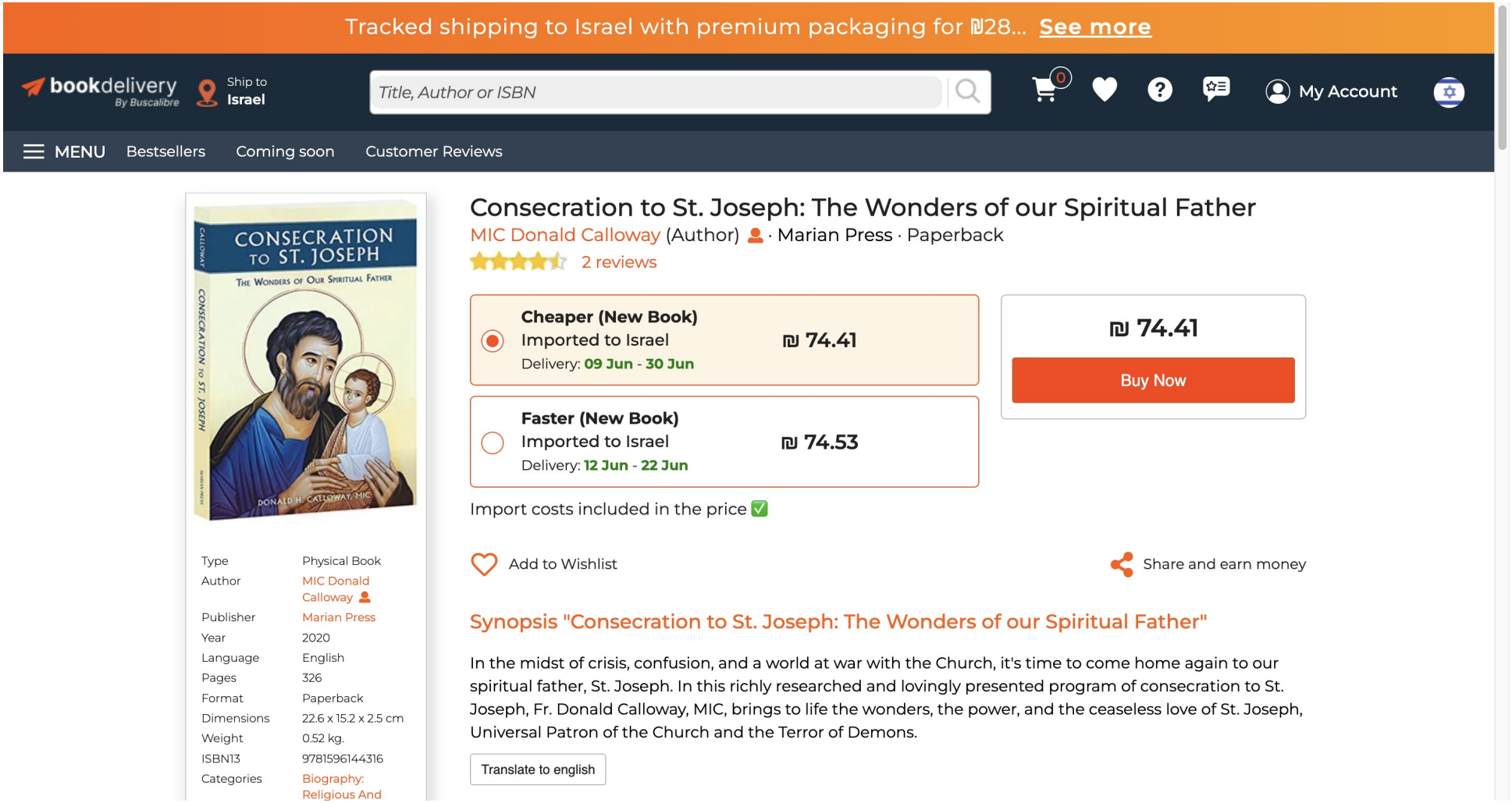
column	mean	std	min	max	median	total_rows
Price in USD	33.03	28.33	7.33	477.52	24.71	3542
Year	2014.21	9.55	1920.00	2026.00	2018.00	3542
StarRating	4.82	0.48	1.00	5.00	5.00	3542
NumberOfReviews	0.95	7.14	0.00	340.00	0.00	3542
NumberOfAuthors	1.62	0.83	0.00	4.00	1.00	3542

Source: *output/books_summary.csv*

6. Output files

File (under output/)	Contents
books_raw.csv	Raw DataFrame, all 20 columns (Step 2)
books_raw.json	Same content as JSON, empty fields stripped (Step 2)
books_example.json	Single example record (Step 2)
books_example.jpg	Screenshot of one of the crawled pages (Step 2)
books_before_sort.csv	First 10 rows before sorting (Step 3)
books_after_sort.csv	First 10 rows after sort by Title ascending (Step 3)
books_processed.csv	Full data + IsExpensive + NumberOfAuthors (Step 4)
books_processed.json	Same content as JSON (Step 4)
books_processed_preview.csv	First 10 rows after processing (Step 4)
books_summary.csv	Summary statistics over the 5 columns (Step 5)
consolidated_report.pdf	This report

7. Crawled-page screenshot (Step 2)



Screenshot of the first book page crawled. Source: *output/books_example.jpg*